Drug Design

Aims:

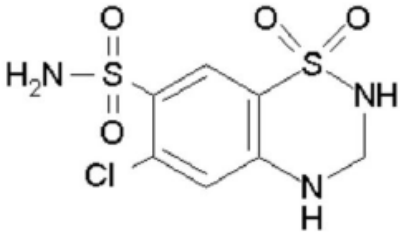
- The goals and objectives of this course will highlight a combination of fundamentals and applications of drug design and development.
- Special intonation will also be placed on the methods used by Medicinal Chemists to design new drugs to treat significant disease states, and by Pharmaceutical Chemists to develop, deliver and monitor effective therapeutic agents.

➢ Definition:

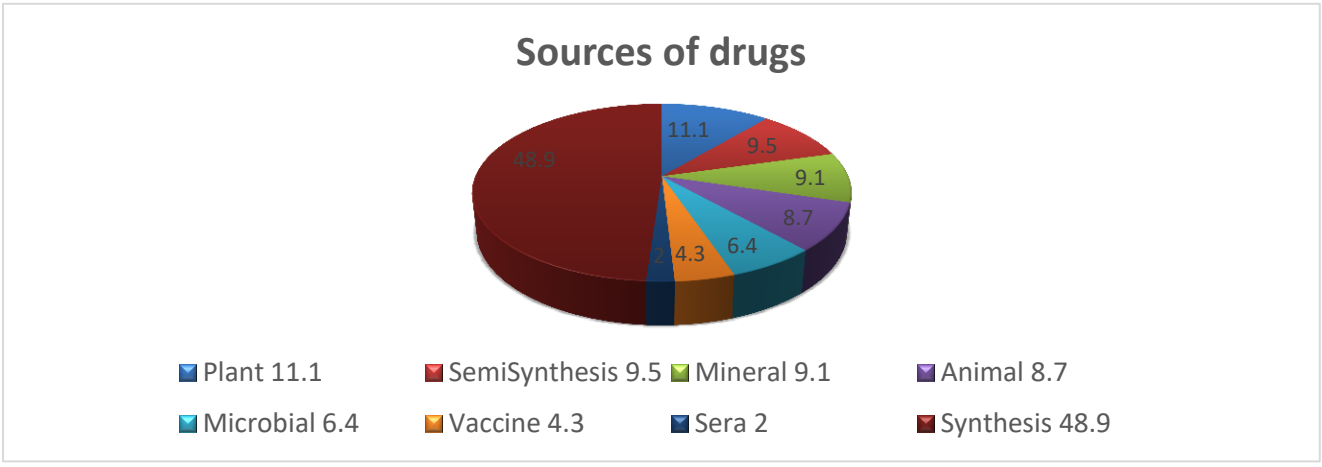
- Drugs are chemical substances that are used to prevent or cure diseases in human, animals and plants.
- The word drug therefore, imposes an action-effect context within which the properties of a substance are described. For example, when a drug is defined as an analgesic, it means that it is used to treat pain .... Thus a drug may have described as having analgesic, vasodepressor, anticonvulsant, antibacterial, ……. etc. properties.

➢ Drug names: (nomenclature)

- Chemical
  - 6-Chloro-3,4-dihydro-7-sulfamoyl-2H-1,2,4-benzothiadiazine 1,1-dioxide
- Trade
  - Hydrodiuril®, Hydroaquil®, Esidrex®, Urozide®, Novohydrazide® etc. Many others
- Generic
  - Hydrochlorothiazide



Sources of drugs:



1. TARGET DISEASE

- Priority for the Pharmaceutical Industry
  Can the profits from marketing a new drug outweigh the cost of developing and testing that drug?

- Questions to be addressed
  - Is the disease widespread? (e.g. cardiovascular disease, ulcers, malaria)
  - Does the disease affect the first world? (e.g. cardiovascular disease, ulcers)
  - Are there drugs already on the market? If so, what are these advantages and disadvantages? (e.g. side effects)
  - Can one identify a market advantage for a new therapy?

- **Choosing a Disease**
  - Pharmaceutical companies are commercial enterprises
  - Pharmaceutical companies tend to avoid products with a small market (i.e. a disease which only affects a small subset of the population)
  - Pharmaceutical companies avoid products that would be consumed by individuals of lower economic status (i.e. a disease which only affects third world countries).
  - Most research is carried out on diseases which afflict “first world" countries: (e.g. cancer, cardiovascular diseases, depression, diabetes, flu, migraine, obesity).

- **Identifying a Drug Target**
  - Drug Target: **macromolecule** or biological system, which the drug will interact with.
  - Sometimes this can happen through incidental observation...

2. Drug Targets:

A) Lipids

(Cell Membrane Lipids)

B) Proteins

(Receptors - Enzymes - Carrier Proteins - Structural Proteins (tubulin))

C) Nucleic Acids

(DNA – RNA)

D) Carbohydrates

(Cell surface carbohydrates - Antigens and recognition molecules)

➢ Target Selectivity

- **Between species**
  - Antibacterial and antiviral agents
  - Identify targets which are unique to the invading pathogen
  - Identify targets which are shared but which are significantly different in structure

- **Within the body**
  - Selectivity between different enzymes, receptors etc.
  - Selectivity between receptor types and subtypes
  - Selectivity between isozymes
  - Organ selectivity

➢ Drug Target Interactions

✓ **Covalent interactions**

✓ **Non-Covalent interactions**

- **Ionic interaction**
- **Hydrogen bonding interactions**
- **Vander Waals interactions**
- **Hydrophobic interaction**

3. Genesis of Drugs

➢ **How are drugs discovered and developed?**

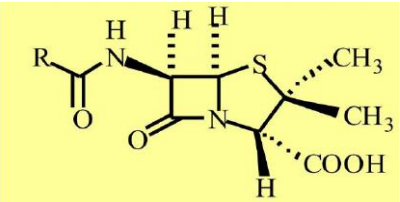
(Lead Structure Discovery)

➢ The Lead Compound

- A compound demonstrating a property likely to be therapeutically useful
- The level of activity and target selectivity are not crucial
- Used as the starting point for drug design and development
- Found by design (molecular modelling or NMR) or by screening compounds (natural or synthetic)
- Need to identify a suitable test in order to find a lead compound
- Active Principle - a compound that is isolated from a natural extract and which is principally responsible for the extract's pharmacological activity. Often used as a lead compound.

➢ Drug Discovery Without a Lead [Serendipity]

- Penicillin's: 1928 - Fleming
- Bacteria lysed by green mold
- mold spore contaminates culture dish
- left dish on bench top while on vacation
- weather was unseasonably cold
- particular strain of mold was a good penicillin Producer

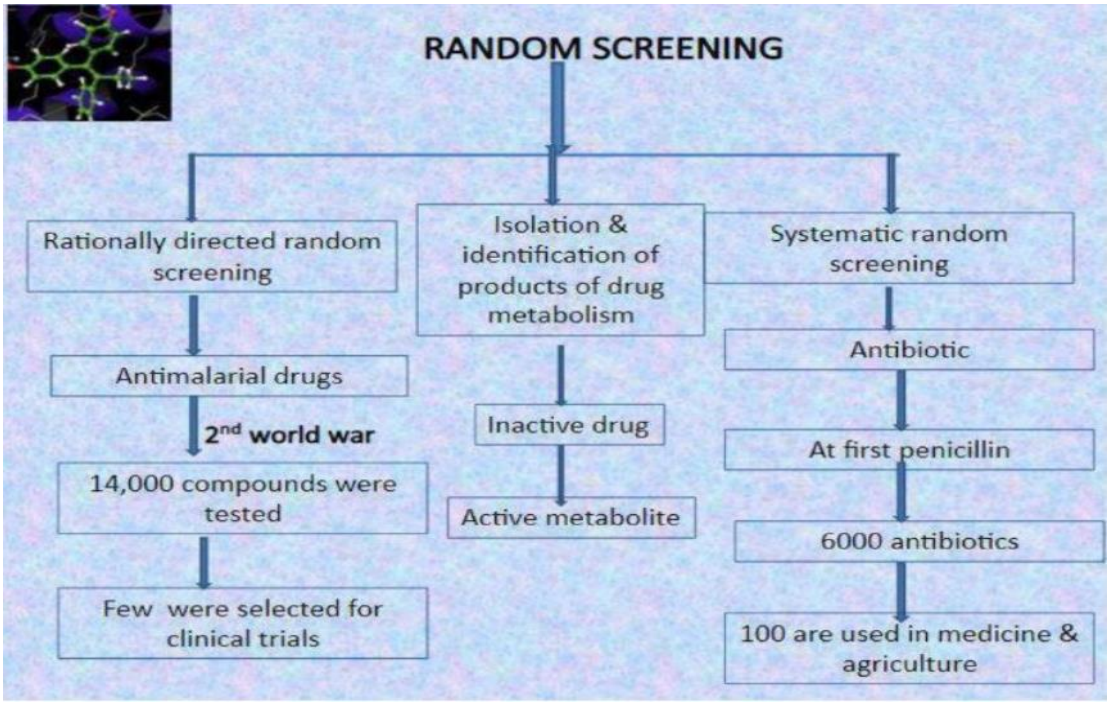


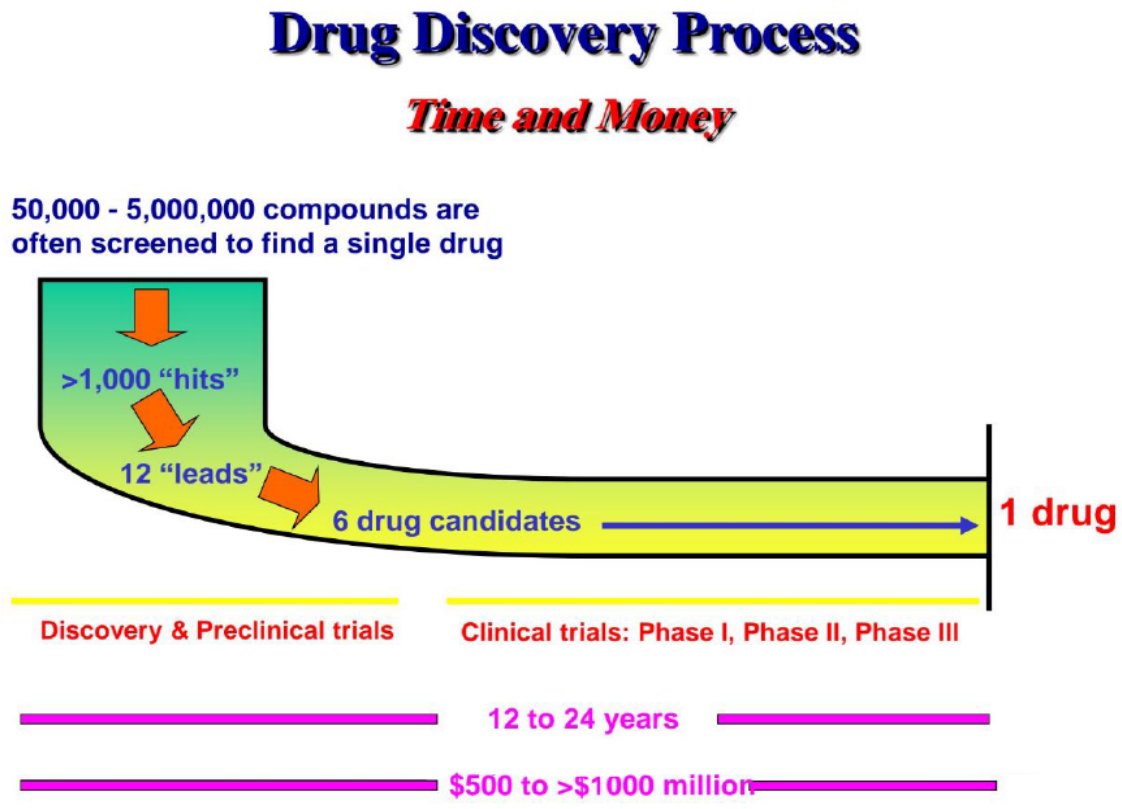
penicillin V (R = $PhOCH_2$)
penicillin G (R = CH_2Ph)
2.2

- Structure of penicillin elucidated in 1944 – x-ray crystal structure

➢ 2. Random Screening

- it is estimated that for a new anticonvulsant discovery through this process it might be necessary to screen about 500,000 chemical compounds





3. 1. Sources of lead components:

A) The Natural world

- ✓ Plant life (flowers, trees, bushes)
- ✓ Micro-organisms (bacteria, fungi)
- ✓ Animal life (frogs, snakes, scorpions)
- ✓ Biochemical (Neurotransmitters, hormones)
- ✓ Marine chemistry (corals, bacteria, fish etc.

B) The Synthetic World

- ✓ Chemical synthesis (traditional)
- ✓ Combinatorial synthesis

C) The Virtual World

- ✓ Computer aided drug design (CADD)

3. 2. Identification of lead components:

A) Isolation and purification

- ✓ Solvent-Solvent Extraction
- ✓ Chromatography
- ✓ Crystallization
- ✓ Distillation

B) Structure determination

- ✓ Elemental Analysis
- ✓ Molecular Weight
- ✓ Mass Spectrum
- ✓ Infra-Red
- ✓ Ultra Violet
- ✓ NMR (1H, 13C, 2D)
- ✓ X-ray Crystallography

3. 3. Lead components from natural world:

Plant extracts

- ✓ **Opium** → Morphine → lead compound for Narcotic analgesic
- ✓ **Cinchona bark** → Quinine → lead compound for anti-malaria
  → Quindine → lead compound for anti-arrhythmic
- ✓ **Yew tree** → Taxol → lead compound for anti-cancer
- ✓ **Cocca** → cocaine → lead compound for local anesthetic
- ✓ **willow bark** → Salicylic acid → lead compound for NSAID